

# Homework 1

## Vectors, Matrices, and Linear Systems

Due Fri, Sep. 18, 2026

You must submit a PDF of your work by 11:59pm on Friday, September 18.

Remember to do all of the following:

- Show all of your work, including brief explanations of what you are doing.
- Keep your work organized and legible, and clearly indicate your final answers.
- If you completed this assignment on paper, use a scanner or scanning app to get a PDF.
- Tag all of the pages each problem appears on when uploading.

Total: 40 points.

### Part 1: Vectors

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Problem 1 (5 pts)

Let  $u = [2, -1, 3]^T$ ,  $v = [1, 4, -2]^T$ , and  $w = [-4, 2, 6]^T$  (column vectors in  $\mathbb{R}^3$ ).

Calculate each of the following.

- (a)  $(2u - v) \cdot w$  1 pt
- (b)  $\|v + (1/2)w\|$  1 pt
- (c) The angle between  $u$  and  $w$  (in degrees, rounded to two decimal places) 3 pts

### Part 2: Matrices

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Problem 2 (6 pts)

Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 0 & -1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix} \quad v = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \quad w = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

Calculate each of the following by hand or explain why they are undefined.

- (a)  $AB$  and  $BA$  1 pt
- (b)  $Av$  and  $vA$  1 pt
- (c)  $Aw$  and  $wA$  1 pt
- (d)  $vw$  and  $wv$  1 pt
- (e)  $A^2$  1 pt
- (f)  $B^2$  1 pt

### Part 3: Linear Systems and Gaussian Elimination

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Problem 3 (6 pts)

Use Gaussian elimination to find the general solution to each linear system.

Show every elementary row operation you use.

(a)  $x_1 + 2x_2 - x_3 = -1$  3  
pts

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 2 \\ -x_1 + x_2 + 2x_3 &= -1 \end{aligned}$$

(b)  $x + 2y - z = 4$  3  
pts

$$\begin{aligned} 2x + 5y + z &= 6 \\ 3x + 7y &= 10 \end{aligned}$$

Problem 4 (8 pts)

A foundry blends three stock alloys to produce 100 kg of a target alloy. The compositions by mass are:

Alloy 1: 50% copper, 30% zinc, 20% nickel

Alloy 2: 20% copper, 70% zinc, 10% nickel

Alloy 3: 40% copper, 20% zinc, 40% nickel

The target alloy must be 36% copper, 44% zinc, and 20% nickel by mass. Assume no metal is lost

in blending, so the mass of each metal going in equals the mass of that metal in the product.

(a) Create a system of three linear equations relating the masses of Alloy 1, Alloy 2, and Alloy 3 used. 3 pts

Explain what any variables you create represent, including units.

(Hint: write one equation for copper, one for zinc, and one for nickel.)

(b) Convert the linear system from part (a) into an augmented matrix. 1 pt

(c) Use Gaussian elimination to find the mass of each alloy needed. 4 pts

Problem 5 (7 pts)

Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

(a) Find the RREF of A, showing all elementary row operations. 2 pts

(b) What is the rank of A? Explain. 1 pt

(c) Depending on the vector b, could the matrix equation  $Ax = b$  have 0, 1, or infinitely many solutions? Explain why each case is possible or impossible. 4 pts

## Part 4: Span and Linear Independence

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Problem 6 (8 pts)

Determine whether or not each set of vectors is linearly independent. Then describe the span of

the vectors geometrically as a point, line, plane, or all of  $\mathbb{R}^3$ .

(a)  $v_1 = [1, 0, 2]^T$ ,  $v_2 = [2, 1, 3]^T$ ,  $v_3 = [0, 1, -1]^T$   
4 pts

(b)  $v_1 = [1, 1, 0]^T$ ,  $v_2 = [0, 1, 1]^T$ ,  $v_3 = [1, 0, 1]^T$   
4 pts